

assignment1

October 5, 2026

1 CS 335 — Fall 2026

1.1 Assignment 1 Submission

Due: Monday, October 5, 2026, 5:00pm

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1.2 Question 1 — (6 marks) No-arbitrage option pricing.

Suppose the current price of a stock is \$18. Consider a two state tree model defined such that after 3 months, the stock price may increase to \$19 with a 40% probability or decrease to \$16 with a 60% probability. Consider a European call option, based on this stock, with strike price $K = 17$ dollars and expiry time $T = 3$ months. Assume the risk-free interest rate is 9%.

1.2.1 (a) — 3 marks

Construct a risk-free portfolio and use it to determine the present no-arbitrage value (fair value) of the option (i.e., at time $t = 0$).

Answer:

Construct the following portfolio Π : - A short position of one call option worth V - A long position of δ shares of the stock

The portfolio value is $\Pi = \delta S - V$.

If stock goes up: - $S = 19$ - $V = 2$ - $\Pi_{up} = 19\delta - 2$

If stock goes down: - $S = 16$ - $V = 0$ - $\Pi_{down} = 16\delta$

$$\Pi_{up} = \Pi_{down} \quad 19\delta - 2 = 16\delta \quad \delta = \frac{2}{3} = .67 \quad \Pi = \frac{32}{3}$$

So the value of the portfolio today is $\Pi_{today} = \Pi e^{-0.09 \times 0.25} = \frac{32}{3} e^{-0.0225}$

But $\Pi_{today} = \delta S_{today} - V_{today}$.

So $V = \delta S_{today} - \Pi_{today} = \frac{2}{3} \times 18 - \frac{32}{3} e^{-0.0225}$

$$V = \$1,57065347$$

1.2.2 (b) — 3 marks

Suppose such an option is priced on the market at $V^* = 2.0$ (i.e., *higher* than the no-arbitrage price). Describe the details for a strategy that will result in a guaranteed profit (in present value, i.e., at $t = 0$) and determine the value of that profit (per option), somewhat similar to what we did in class. (As usual in this course, we will assume short-selling is allowed and the no-arbitrage principle holds.)

Answer:

1. Sell an option at market price $V^* = 2.0$.
2. Borrow \$10 from the bank at the risk free interest rate. We now have \$12 cash.
3. Purchase δ shares of the stock, which right now cost \$12
4. After 3 months, our portfolio is either worth 16δ , or $19\delta - 2$ (because we have to pay out the option). Either way, this is equal to a value of $\Pi = \frac{32}{3}$.
5. Sell the portfolio for $\frac{32}{3}$ cash.
6. We have to pay back the bank $10e^{0.0225}$.
7. Our profit is $\frac{32}{3} - 10e^{0.0225} = \439116325

1.3 Question 2 — (10 marks) Coding: Convergence of a walk on a discrete lattice.

Consider a random walk on an N step lattice on $t \in [0, T]$ with the parameters

$$\begin{aligned} X_{n+1} &\rightarrow X_n + \sigma\sqrt{\Delta t} ; && \text{with probability } p \\ X_{n+1} &\rightarrow X_n - \sigma\sqrt{\Delta t} ; && \text{with probability } 1 - p \\ p &= \frac{1}{2} \left[1 + \frac{\alpha}{\sigma} \sqrt{\Delta t} \right] \\ \Delta t &= \frac{T}{N}, \end{aligned}$$

which converges to the solution of the SDE

$$dX = \alpha dt + \sigma dZ,$$

as $\Delta t \rightarrow 0$. The exact solution to this SDE is such that the exact density (probability distribution) of $X(T)$ is a normal density with mean $X_{init} + \alpha T$ and standard deviation $\sigma\sqrt{T}$, assuming the initial value of X is X_{init} .

Verify this experimentally via numerical simulation of many lattice walks, by preparing a Jupyter/Python notebook to carry out the following. Perform a lattice walk simulation, using the parameters given in Table 1. Starting at $X_0 = X_{init}$, the value of X is then moved up with probability p or down with probability $(1 - p)$. A *uniformly* distributed random number on $[0, 1]$ should be generated and used to determine the actual move taken. At the next lattice node (i.e., time step), this process is repeated, until we obtain the final position of X after N steps, X_N , at $t = T$. This is a single simulation outcome; we perform many such simulations to produce many (possibly) different outcomes. Because each step moves up or down a fixed amount, the set of all possible outcomes for this discrete lattice walk is

$$X_N^j = X_{init} + \{(2j - N)\sigma\sqrt{\Delta t}\} \quad j = 0, \dots, N.$$

As indicated in Table 1, you should perform the above process first for 1000 simulations, then separately for 10,000 simulations, and finally for 100,000 simulations. In each case, print out the number of simulations and the corresponding mean and standard deviation of the computed values of X_N from the simulations in that set. Also, print out the anticipated exact mean and standard deviation for comparison.

Next, for each set of simulations, we would like to generate a histogram of the observed *probability density* versus $X(T)$ as described below.

Care must be taken in scaling this histogram to get the correct probability density. Let $p(x)$ be the probability density, and let $\text{bin}_{[a,b]}$ be the bin on an interval $[a, b]$ of the x axis. Then

$$\begin{aligned} \int_a^b p(x) dx &\simeq p((a+b)/2)(b-a) \\ &\simeq \frac{\text{Number of occurrences in bin}_{[a,b]}}{\text{Total number of occurrences}} \end{aligned}$$

so that

$$p((a+b)/2) \simeq \frac{1}{(b-a)} \frac{\text{Number of occurrences in bin}_{[a,b]}}{\text{Total number of occurrences}}.$$

Use $N + 1$ bins, with the bin boundaries falling half-way between the lattice nodes at timestep N . Use matplotlib's *bar* function to display your probability density plot constructed as described above. (Alternatively, if you prefer, Matplotlib's *hist* or Numpy's *histogram* functionality can automate some of this process, if you give it appropriately chosen values for the range, bin width, and density parameters.) Be sure to include appropriate labeling on your plots (axis labels, title, legend if necessary).

Then, for comparison, over top of each of your histograms, generate a standard (line) plot of the true normal distribution with mean $X_{init} + \alpha T$ and standard deviation $\sigma\sqrt{T}$, over the same range (in a different color). The Scipy function `stats.norm.pdf` can be used to evaluate a true normal distribution function. Thus each of your three plots will look something roughly along the lines of Figure 1 (albeit generated with different data).

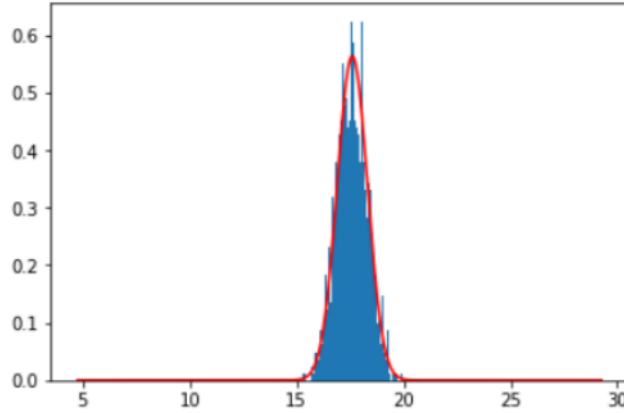


Figure 1: A histogram of an approximate probability density function, with a normal distribution plotted on top.

Parameter	Value
T	1.75
σ	0.3
α	0.4
X_{init}	17
Number of simulations	1000, 10000, 100000
Number of timesteps (N)	300

Your code must be *vectorized* for efficiency, i.e., prefer vector operations instead of scalar operations. As a very simple example of vectorization, if I have a Numpy vector V and I want to add 1 to every entry of the vector, I can simply write the vector operation $V = V + 1$, instead of using a slower for loop over index i and separately computing $V[i] = V[i] + 1$ for each entry. Similarly, many Numpy functions work on vectors rather than single numbers.

Specifically, your code for the lattice walk should only explicitly loop over the timesteps, not over the individual simulations - for each time step, the updates for all the simulations (within a set) should be computed together using vector operations. Also, you should not store the entire lattice at once, but rather only one or two columns of it, to save on memory. The example lattice walk code in the course notes is likely to be *very helpful*, since it illustrates both vectorization and lattice walk implementation! (Course notes, page 17).

In addition to the NumPy functions mentioned previously above, you may also find the *mean*, *std*, and `random.uniform` functions useful.

In a markdown cell in your Jupyter notebook, describe what you observe about your results (plots and statistics) as the number of simulations increases from 1000 to 100,000.

1.3.1 Answer — code and results

```
[103]: import numpy as np

T = 1.75
sigma = 0.3
alpha = 0.4
X_init = 17.0
N = 300
sims = [1000,10000,100000]

dt = T / N
p = (1 / 2) * (1 + np.sqrt(dt) * alpha / sigma)
dh = sigma * np.sqrt(dt)

anticipated_mean = X_init + alpha * T
anticipated_std = sigma * np.sqrt(T)

xs = [np.full(sims[i], X_init) for i in range(len(sims))]

[104]: for i in range(len(sims)):
    ts = np.linspace(0, T, N)
    for t in ts:
        xs[i] += np.random.choice([dh,-dh],sims[i],p=[p,(1-p)])
    simulated_mean = np.mean(xs[i])
    simulated_std = np.std(xs[i])

    print(f"Run {i+1} ({sims[i]} simulations): ")
    print(f"\t- Anticipated mean: {anticipated_mean:.3f} \t Anticipated std:␣
↪{anticipated_std:.3f}")
    print(f"\t- Simulated mean: {simulated_mean:.3f} \t Simulated std:␣
↪{simulated_std:.3f}")
```

Run 1 (1000 simulations):

- Anticipated mean: 17.700	Anticipated std: 0.397
- Simulated mean: 17.682	Simulated std: 0.395

Run 2 (10000 simulations):

- Anticipated mean: 17.700	Anticipated std: 0.397
- Simulated mean: 17.702	Simulated std: 0.393

Run 3 (100000 simulations):

- Anticipated mean: 17.700	Anticipated std: 0.397
- Simulated mean: 17.699	Simulated std: 0.395

```
[ ]: import matplotlib.pyplot as plt
from scipy.stats import norm

fig, axs = plt.subplots(1,len(sims), figsize=(6 * len(sims), 5))
```

```

min_X = X_init - N * dh
max_X = X_init + N * dh

x = np.linspace(min_X, max_X, 1000)
y = norm.pdf(x, loc=anticipated_mean, scale=anticipated_std)

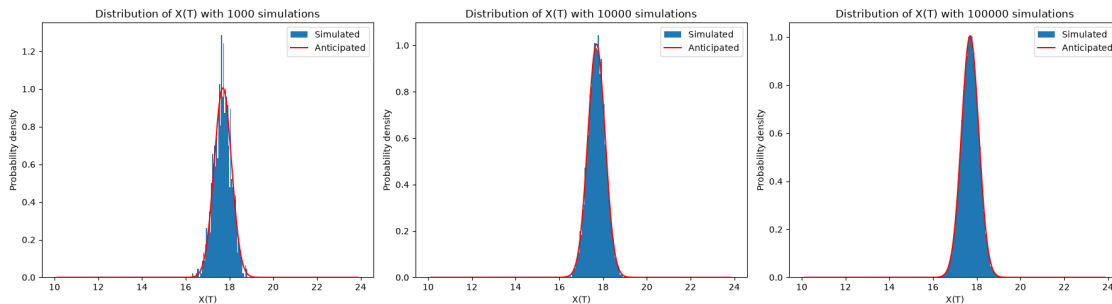
bin_edges = np.linspace( min_X - dh, max_X + dh, N + 2)

for i in range(len(sims)):
    axs[i].set_title(f"Distribution of X(T) with {sims[i]} simulations")
    axs[i].set_xlabel("X(T)")
    axs[i].set_ylabel("Probability density")

    axs[i].hist(xs[i], bins=bin_edges, density = True, label = "Simulated")
    axs[i].plot(x,y,"r",label="Anticipated")

    axs[i].legend(loc="upper right")
plt.tight_layout()
plt.show()

```



1.3.2 Answer — observations

For the statistics regarding the simulated mean and std vs the anticipated ones, we observe that even with 1000 simulations we get results that are already within $\pm 5\%$ of the anticipated ones. However, as the number of simulations increase, our results are even closer.

Same idea for the plots basically. As the number of simulations increase, our simulated values get close and closer to representing the exact distribution

1.4 Question 3 — (6 marks) Coding: Brownian motion and Ito calculus.

Let

$$\begin{aligned}\Delta Z_i &= \phi(t_i) \sqrt{\Delta t} \\ \phi(t_i) &\sim N(0, 1) \\ Z(t_0 = 0) &= 0 \\ Z(t_i) &= \sum_{j=0}^{i-1} \Delta Z_j \\ t_i &= i \Delta t\end{aligned}$$

and

$$\begin{aligned}I(T, \Delta t) &= \sum_{i=0}^{N-1} (\Delta Z_i)^2 \\ \Delta t &= \frac{T}{N}.\end{aligned}$$

Write a Jupyter/Python notebook to evaluate $I(T, \Delta t)$. For $T = 2$, for fixed Δt , compute the value of $I(T, \Delta t)$. Use 70000 paths, and compute the mean and variance of the result, for $N = 100, 200, 400, 800, 1600, 3200, 6400$ timesteps. Vectorize your code so that there is only one explicit for loop over the timesteps, for a given value of N .

Write code to generate a simple text-based table of mean and variance of $I(T, \Delta t)$ for the different values of Δt . (No fancy formatting is required! You can just use Python's print function to generate each row, with appropriate arguments.)

Plot a graph of $\text{var}(I(T, \Delta t))$ versus Δt , with appropriate labels. In a markdown cell of your Jupyter notebook, explain mathematically what you observe.

1.4.1 Answer — code and results

```
[13]: import numpy as np
import matplotlib.pyplot as plt

T = 2
paths = 70000
Ns = np.array([100 * 2**i for i in range(7)])

dts = T / Ns

means = []
vars = []
Is = []
table_data = []
```

```
[14]: for i,N in enumerate(Ns):
        I = np.zeros(paths)
        for j in range(N):
            I += np.square(np.random.normal(size=paths) * np.sqrt(dts[i]))
        means.append(np.mean(I))
        vars.append(np.var(I))
        Is.append(I)
        table_data.append([
            f"{N}",
            f"{dts[i]:.6f}",
            f"{means[i]:.6f}",
            f"{vars[i]:.6f}"
        ])
    ])
```

```
[25]: fig, axs = plt.subplots(1,2,figsize=(12,5))

axs[0].axis("off")

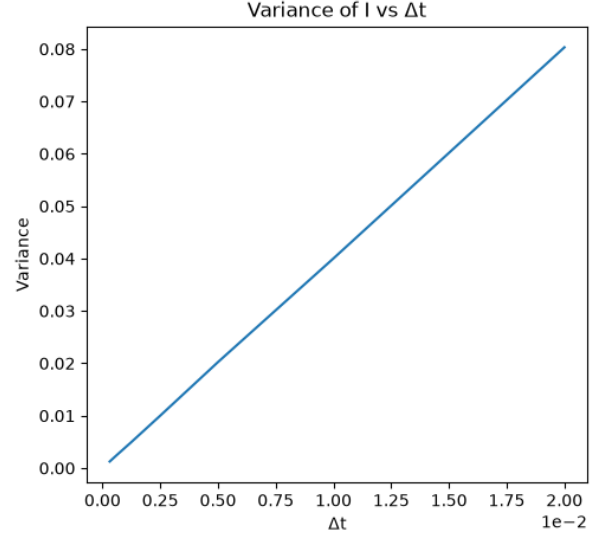
table = axs[0].table(
    cellText=table_data,
    colLabels=["N", " $\Delta t$ ", "Mean", "Variance"],
    loc="center"
)
table.scale(1, 1.5)

axs[1].plot(dts, vars)
axs[1].set_xlabel(" $\Delta t$ ")
axs[1].set_ylabel("Variance")
axs[1].set_title("Variance of I vs  $\Delta t$ ")

axs[1].ticklabel_format(
    axis="x",
    style="sci",
    scilimits=(0,0)
)

plt.show()
```

N	Δt	Mean	Variance
100	0.020000	1.998491	0.080320
200	0.010000	2.000791	0.040023
400	0.005000	2.000238	0.020233
800	0.002500	2.000310	0.010008
1600	0.001250	1.999577	0.004963
3200	0.000625	1.999869	0.002509
6400	0.000313	1.999822	0.001249



1.4.2 Answer — observations and mathematical explanation

So we have

$$\Delta Z_i = \phi \sqrt{\Delta t}$$

And

$$(\Delta Z_i)^2 = (\phi \sqrt{\Delta t})^2$$

So we can now look at

$$E[(\Delta Z_i)^2] = E[(\phi \sqrt{\Delta t})^2] = \Delta t E[\phi^2]$$

$$Var[(\Delta Z_i)^2] = Var[(\phi \sqrt{\Delta t})^2] = \Delta t^2 Var[\phi^2]$$

And we know that since ϕ is normal, we have $E[\phi^2] = 1$ and $Var[\phi^2] = 2$

So

$$E[(\Delta Z_i)^2] = \Delta t$$

$$Var[(\Delta Z_i)^2] = 2\Delta t^2$$

Also,

$$\Delta t = \frac{T}{N}$$

Now, we can look at $I(T, \Delta t) = \sum_{i=0}^{N-1} (\Delta Z_i)^2$

$$E[I(T, \Delta t)] = E\left[\sum_{i=0}^{N-1} (\Delta Z_i)^2\right] = N \times E[(\Delta Z_i)^2] = N \times \Delta t = T = 2$$

This checks out from our experimental data, where our calculated mean averages to around ~ 2 .

$$Var[I(T, \Delta t)] = Var\left[\sum_{i=0}^{N-1} (\Delta Z_i)^2\right] = N \times Var[(\Delta Z_i)^2] = N \times 2\Delta t^2 = 2T\Delta t = 4\Delta t$$

Which goes to 0 as Δt goes to 0. This also checks out from our experimental data and graph.

1.5 Question 4 — (5 marks) Ito calculus.

1.5.1 (a) — 3 marks

Let $X(t)$ and $Y(t)$ be two stochastic processes, such that

$$\begin{aligned} dX &= \mu_X(X(t), t)dt + \sigma_X(X(t), t)dZ_X \\ dY &= \mu_Y(Y(t), t)dt + \sigma_Y(Y(t), t)dZ_Y \end{aligned}$$

with $dZ_X(t)$, $dZ_Y(t)$ being the increments for two distinct Wiener processes, $Z_X(t)$ and $Z_Y(t)$. Let $X(t_i) = X_i$ and $Y(t_i) = Y_i$. Show that

$$(X_{i+1} - X_i)(Y_{i+1} - Y_i) = X_{i+1}Y_{i+1} - X_iY_i - X_i(Y_{i+1} - Y_i) - Y_i(X_{i+1} - X_i).$$

Then, using the definition of the Ito integral which is the limit of a discrete sum, show that

$$\int_0^t X(s)dY(s) = [XY]_0^t - \int_0^t Y(s)dX(s) - \int_0^t dX(s)dY(s).$$

Answer:

Let

$$\Delta X_i = X_{i+1} - X_i, \quad \Delta Y_i = Y_{i+1} - Y_i.$$

So,

$$\begin{aligned} \Delta X_i \Delta Y_i &= X_{i+1}Y_{i+1} - X_i(Y_i + \Delta Y_i) - Y_i(X_i + \Delta X_i) + Y_iX_i \\ \Delta X_i \Delta Y_i &= X_{i+1}Y_{i+1} - X_i\Delta Y_i - Y_i\Delta X_i - Y_iX_i \\ (X_{i+1} - X_i)(Y_{i+1} - Y_i) &= X_{i+1}Y_{i+1} - X_iY_i - X_i(Y_{i+1} - Y_i) - Y_i(X_{i+1} - X_i). \end{aligned}$$

Which is what we were asked to show.

Also note that

$$X_i \Delta Y_i = X_{i+1}Y_{i+1} - Y_iX_i - Y_i\Delta X_i - \Delta X_i \Delta Y_i$$

Now,

$$\int_0^t X(s)dY(s) = \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} X_i \Delta Y_i$$

Substituting $X_i \Delta Y_i$, we get

$$\int_0^t X(s)dY(s) = \lim_{N \rightarrow \infty} \sum_{i=0}^{N-1} X_{i+1}Y_{i+1} - Y_iX_i - Y_i\Delta X_i - \Delta X_i \Delta Y_i$$

Note that $\sum_{i=0}^{N-1} X_{i+1}Y_{i+1} - Y_iX_i = X_NY_N - X_0Y_0 = [XY]_0^t$

So,

$$\begin{aligned}\int_0^t X(s) dY(s) &= [XY]_0^t + \lim_{N \rightarrow \infty} \left(\sum_{i=0}^{N-1} -Y_i \Delta X_i - \Delta X_i \Delta Y_i \right) \\ \int_0^t X(s) dY(s) &= [XY]_0^t - \int_0^t Y(s) dX(s) - \int_0^t dX(s) dY(s)\end{aligned}$$

1.5.2 (b) — 2 marks

Let $Z(t)$ be a stochastic process satisfying $dZ = \phi\sqrt{dt}$, and assume $Z(0) = 0$. Using the result in part (a), show that (assuming Ito calculus)

$$\int_0^t Z(s) dZ(s) = \frac{1}{2} Z(t)^2 - \frac{t}{2}.$$

Answer:

$$\begin{aligned}\int_0^t Z(s) dZ(s) &= [ZZ]_0^t - \int_0^t Z(s) dZ(s) - \int_0^t dZ(s) dZ(s) \\ \int_0^t Z(s) dZ(s) &= Z(t)^2 - Z(0)^2 - \int_0^t Z(s) dZ(s) - \int_0^t dZ(s) dZ(s) \\ 2 \int_0^t Z(s) dZ(s) &= Z(t)^2 - \int_0^t dZ^2(s) \\ \int_0^t Z(s) dZ(s) &= \frac{1}{2} Z(t)^2 - \frac{1}{2} \int_0^t dZ^2(s) \\ \int_0^t Z(s) dZ(s) &= \frac{1}{2} Z(t)^2 - \frac{1}{2} \int_0^t dZ^2(s) \\ \int_0^t Z(s) dZ(s) &= \frac{1}{2} Z(t)^2 - \frac{t}{2}.\end{aligned}$$

1.6 Question 5 — (10 marks) Ito calculus and Ito's Lemma.

Suppose dZ is the increment of a Wiener process and

$$dX = \mu(X(t), t)dt + \sigma(X(t), t)dZ.$$

1.6.1 (a) — 6 marks

If

$$F(X, t) = X^4 e^{-t} + 3X \cos t,$$

what is $dF(X, t)$ (interpreted in the sense of Ito calculus)?

Answer: From Ito's lemma:

$$\begin{aligned} dF(X, t) &= F_t dt + F_X dX + \frac{1}{2} F_{XX} (dX)^2 \\ &= (\mu F_X + \frac{\sigma^2}{2} F_{XX} + F_t) dt + \sigma F_X dZ \end{aligned}$$

now, to compute the partial derivatives:

$$\begin{aligned} F_X &= 4X^3 e^{-t} + 3 \cos(t) \\ F_{XX} &= 12X^2 e^{-t} \\ F_t &= -X^4 e^{-t} - 3X \sin(t) \end{aligned}$$

Substituting them back in

$$\begin{aligned} dF(X, t) &= (\mu(4X^3 e^{-t} + 3 \cos(t)) + \frac{\sigma^2}{2} 12X^2 e^{-t} - X^4 e^{-t} - 3X \sin(t)) dt + \sigma(4X^3 e^{-t} + 3 \cos(t)) dZ \\ &= (\mu(4X^3 e^{-t} + 3 \cos(t)) + 6\sigma^2 X^2 e^{-t} - X^4 e^{-t} - 3X \sin(t)) dt + \sigma(4X^3 e^{-t} + 3 \cos(t)) dZ \end{aligned}$$

1.6.2 (b) — 4 marks

Using dX as defined above, and expanding, simplify

$$\int_0^T (dX)^2$$

as much as possible. The integral is to be interpreted in the Ito calculus sense. (Note that $\sigma(X(t), t)$ is a function of t , not a constant, so you will not be able to fully solve the integral.)

Answer:

$$\begin{aligned} \int_0^T (dX)^2 &= \int_0^T (\mu(X(t), t) dt + \sigma(X(t), t) dZ)^2 \\ &= \int_0^T \mu^2 dt^2 + 2\mu dt \sigma dZ + \sigma^2 dZ^2 \\ &= \int_0^T \mu^2 dt^2 + 2 \int_0^T \mu dt \sigma dZ + \int_0^T \sigma^2 dZ^2 \end{aligned}$$

But in Ito calculus we have $dt^2 = 0$, $dt dZ = 0$ and $dZ^2 = dt$, so:

$$\int_0^T \mu^2 dt^2 = 0$$

and

$$\int_0^T \mu \sigma dt dZ = 0$$

Therefore,

$$\begin{aligned} \int_0^T (dX)^2 &= \int_0^T \sigma^2 dt \\ &= \int_0^T \sigma(X(t), t)^2 dt \end{aligned}$$